The trace identity $D = \mathrm{cross}_{(3,1,1)\to(3,2)}$: restatement as a three-way KL cross-trace identity and the S_4-double-branching framework

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Abstract

Note: this writeup overlaps substantially with the parallel May 7 paper `2026-05-07-trace-identity.tex`, which gave the cleanest convention-corrected reduction of the identity $D(q) = \mathrm{cross}^{KL}_{(3,1,1)\to(3,2)}(q)$ to a residual cross $= (1+q)(T_A + T_B)$. The distinctive contribution of the present paper is two-fold: (i) the 3-cross-trace identity restatement $\mathrm{cross}^{KL}_{(2,2,1)\to(2,2,1)} + \mathrm{cross}^{KL}_{(3,1,1)\to(3,1,1)} = \mathrm{cross}^{KL}_{(3,1,1)\to(3,2)}$, which casts the surprise as a relation purely among KL-block cross-traces of $R_5\Pi$; (ii) the explicit S_4-double-branching framework that links T_A, T_B to the isotypic structure $V_{(3,2,1)} \downarrow S_4 = 2V_{(3,1)} \oplus 2V_{(2,2)} \oplus 2V_{(2,1,1)}$ with Hoefsmit's seminormal formulas.

The companion writeup `2026-05-06-paths-explicit-and-Q3-refuted.tex` discovered an exact and unexplained identity $D(q) = \mathrm{cross}^{KL}_{(3,1,1)\to(3,2)}(q)$ relating two completely different decompositions of $\sigma_1(V_{(3,2,1)})$: $D(q)$ is the trace contribution where step 11 (the s_5 step) of the staircase Bott–Samelson acts as $(1+q)D_5$, while $\mathrm{cross}^{KL}_{(3,1,1)\to(3,2)}$ is the KL-basis cross-block trace from the $V_{(3,1,1)}$-summand to the $V_{(3,2)}$-summand under branching $V_{(3,2,1)} \downarrow S_5$. This note gives a partial structural reduction. We prove (cleanly, structurally):

$$D(q) \;=\; (1+q)\cdot \mathrm{tr}\big(P_{\mathrm{ascents}_5}\, R_5'\, \Pi_q^{S_5}\big) \;=\; (1+q)(X+Y),$$

where $X = \mathrm{tr}(P_{(2,2,1)} R_5'\Pi)$ and $Y = \mathrm{tr}(P_{\{1,3,9\}} R_5'\Pi)$ are the diagonal traces of $R_5'\Pi$ on the two parts of the s_5-ascent set. Together with computational verification, this identifies

$$D(q) \;\overset{\text{numerically}}{=}\; \mathrm{cross}^{KL}_{(2,2,1)\to(2,2,1)}(R_5\Pi) \;+\; \mathrm{cross}^{KL}_{(3,1,1)\to(3,1,1)}(R_5\Pi),$$

restating the trace identity as a three-way KL cross-trace relation

$$\mathrm{cross}^{KL}_{(2,2,1)\to(2,2,1)} + \mathrm{cross}^{KL}_{(3,1,1)\to(3,1,1)} \;=\; \mathrm{cross}^{KL}_{(3,1,1)\to(3,2)}.$$

The structural proof is partial: the $V_{(2,2,1)}$ piece is fully structural modulo a single vanishing identity (V3), but the $V_{(3,1,1)}$ piece requires an additional structural identity (V_Y) linking the M_5-mediated coupling within $V_{(3,1,1)}$ to the R_5'-cross-flow between $\{1,3,9\}$ and $V_\lambda \setminus V_{(3,1,1)}$. We give the S_4-double-branching framework that suggests where these identities come from.

1 Setup

We adopt the framework of `2026-04-24-sigma1-G1.tex` and the data conventions of `2026-05-06-paths-explicit`
which we summarize.

Hecke algebra and KL basis

Let $H_q = H_q(\mathfrak{S}_n)$ with $T_i^2 = (q-1)T_i + q$ and $q = v^2$. For $\lambda \vdash n$ let V_λ be the cell module with KL basis $\{C_w\}$ indexed by SYTs of shape λ (via RSK to a fixed left cell $\mathcal{C}_\lambda$). In the convention used here (consistent with the implementation in `~/projects/scratch/2026-05-06-paths/wgraph_fast.py`),

$$T_i + 1 \;=\; (1+q)\, D_i + v\, M_i,$$

where D_i is the diagonal $\{0,1\}$-matrix with $(D_i)_{ww} = [s_i \notin D_L(w)]$ (indicator that s_i is a left ascent of w), and M_i has rows in s_i-descent vertices and columns in s_i-ascent vertices, with entries the KL μ-coefficients.

In particular, $T_i + 1$ has zero descent columns: the action of $T_i + 1$ on descent vertices kills them.

The case $\lambda = (3,2,1) \vdash 6$

The cell $\mathcal{C}_{(3,2,1)}$ has 16 SYTs. Branching $V_{(3,2,1)} \downarrow S_5$ into μ-blocks (indexed by the position of 6 in the SYT):

- $V_{(3,2)}$-block: SYTs with 6 in row 2. Indices $\{0,2,4,8,10\}$. 5 vertices.
- $V_{(3,1,1)}$-block: SYTs with 6 in row 1. Indices $\{1,3,6,9,12,14\}$. 6 vertices.
- $V_{(2,2,1)}$-block: SYTs with 6 in row 0. Indices $\{5,7,11,13,15\}$. 5 vertices.

The s_5-descent / s_5-ascent partition (where "descent" means 5 is in a strictly lower row than 6 in the SYT — wait, in our convention, "descent" means $5 \in D_L$, equivalently 6 is in a strictly higher row than 5, equivalently the box of 6 has a smaller row index than the box of 5, which means 5 is "below or beside" 6):

- s_5-ascents ($= D_5$-support): $V_{(2,2,1)} \cup \{1,3,9\}$. 8 vertices.
- s_5-descents ($= D_5$ kernel): $V_{(3,2)} \cup \{6,12,14\}$. 8 vertices.

Within $V_{(3,1,1)}$-block, the descent subset $\{6,12,14\}$ corresponds to SYTs where 5 is in row 0 (above the row-1 position of 6), i.e., the $V_{(2,1,1)}$-double-branching descent copies. The ascent subset $\{1,3,9\}$ corresponds to SYTs where 5 is in row 2 (below 6), i.e., the $V_{(3,1)}$-double-branching ascent copies.

Operators and decompositions

The S_6-staircase reduced word is $(1;2,1;3,2,1;4,3,2,1;5,4,3,2,1)$ of length 15. Define

$$\begin{aligned}
\Pi_q^{S_5} &:= \prod_{j=1}^{10} (T_{i_j} + 1) \quad \text{(staircase Bott–Samelson for } S_5\text{)},\\
R_5' &:= (T_4+1)(T_3+1)(T_2+1)(T_1+1) \quad \text{(steps 12–15)},\\
R_5 &:= (T_5+1)R_5' = (1+q)D_5R_5' + vM_5R_5'.
\end{aligned}$$

Then $\sigma_1(V_{(3,2,1)}) = \mathrm{tr}(\Pi_q^{S_5} R_5)$. Splitting the s_5-step into its diagonal and off-diagonal pieces gives the "Q3 split":

$$\begin{aligned}
\sigma_1 &= D + G,\\
D(q) &:= (1+q) \cdot \mathrm{tr}(\Pi_q^{S_5} D_5 R_5'),\\
G(q) &:= v \cdot \mathrm{tr}(\Pi_q^{S_5} M_5 R_5').
\end{aligned}$$

The KL-basis cross-trace is, for $\mu, \nu \subset \lambda$:

$$\mathrm{cross}^{KL}_{\mu\to\nu}(R_5\,\Pi_q^{S_5}) \;:=\; \mathrm{tr}\big(P_\nu R_5 P_\mu\,\Pi_q^{S_5}\big) \;=\; \sum_{i\in V_\mu,\, j\in V_\nu} (\Pi_q^{S_5})_{ij}\,(R_5)_{ji},$$

where P_μ is the diagonal projector onto V_μ in the KL basis.

By direct enumeration in `~/projects/scratch/2026-05-06-paths/` and in the trace computations of `~/projects/scratch/2026-05-06-trace-identity/` (May 6, evening):

$$\begin{aligned}
D(q) &= q^5 + 8q^6 + 19q^7 + 19q^8 + 8q^9 + q^{10} = q^5(1+q)(q^4+7q^3+12q^2+7q+1),\\
G(q) &= q^4(1+q)(q^6+14q^5+59q^4+94q^3+59q^2+14q+1),\\
\mathrm{cross}^{KL}_{(3,1,1)\to(3,2)}(R_5\Pi) &= q^5 + 8q^6 + 19q^7 + 19q^8 + 8q^9 + q^{10} = D(q),\\
\mathrm{cross}^{KL}_{(2,2,1)\to(2,2,1)}(R_5\Pi) &= q^6 + 3q^7 + 3q^8 + q^9 = q^6(1+q)^3,\\
\mathrm{cross}^{KL}_{(3,1,1)\to(3,1,1)}(R_5\Pi) &= q^5 + 7q^6 + 16q^7 + 16q^8 + 7q^9 + q^{10} = q^5(1+q)^3(1+4q+q^2),\\
\mathrm{cross}^{KL}_{(2,2,1)\to(3,2)}(R_5\Pi) &= q^6 + 4q^7 + 4q^8 + q^9 = q^6(1+q)(1+3q+q^2).
\end{aligned}$$

The other four off-diagonal cross-block traces ($\mathrm{cross}_{(3,2)\to *}$, $\mathrm{cross}_{(3,1,1)\to(2,2,1)}$, $\mathrm{cross}_{(2,2,1)\to(3,1,1)}$) are identically zero; the cross-flux flows entirely *into* $V_{(3,2)}$.

2 The structural decomposition of $D(q)$

Step 1: D as an ascent-restricted diagonal trace

Lemma 1 (Restriction of D to ascents)**.**

$$D(q) \;=\; (1+q)\cdot\Big[\mathrm{tr}\big(P_{(2,2,1)}\,R_5'\,\Pi_q^{S_5}\big) \;+\; \mathrm{tr}\big(P_{\{1,3,9\}}\,R_5'\,\Pi_q^{S_5}\big)\Big] \;=\; (1+q)\,(X+Y),$$

where $X := \mathrm{tr}(P_{(2,2,1)} R_5' \Pi_q^{S_5})$ *and* $Y := \mathrm{tr}(P_{\{1,3,9\}} R_5' \Pi_q^{S_5})$.

Proof. By definition $D(q) = (1+q)\mathrm{tr}(\Pi_q^{S_5} D_5 R_5') = (1+q)\mathrm{tr}(D_5 R_5' \Pi_q^{S_5})$ (cyclicity). The matrix D_5 is the diagonal projector onto s_5-ascent vertices, which by the SYT analysis above is the set $V_{(2,2,1)} \cup \{1,3,9\}$. So $D_5 = P_{V_{(2,2,1)}\cup\{1,3,9\}} = P_{(2,2,1)} + P_{\{1,3,9\}}$ (orthogonal sum of two disjoint projectors). Linearity of trace gives the claim. $\square$

Numerically (`compute_traces.py`, May 6 evening):

$$\begin{aligned}
X &= q^6 + 2q^7 + q^8 = q^6(1+q)^2,\\
Y &= q^5 + 6q^6 + 10q^7 + 6q^8 + q^9 = q^5(1+q)^2(1+4q+q^2),\\
(1+q)(X+Y) &= q^5 + 8q^6 + 19q^7 + 19q^8 + 8q^9 + q^{10} = D(q). \quad \checkmark
\end{aligned}$$

Step 2: identifying $(1+q)X$ as a cross-block trace

Lemma 2 (Diagonal cross-trace at $V_{(2,2,1)}$)**.**

$$\mathrm{cross}^{KL}_{(2,2,1)\to(2,2,1)}(R_5\,\Pi_q^{S_5}) \;=\; (1+q)\,X.$$

Proof. Expand $R_5 = (1+q)D_5R_5' + vM_5R_5'$ and use linearity:

$$\begin{aligned}\operatorname{cross}^{KL}_{(2,2,1)\to(2,2,1)}(R_5\Pi) &= \operatorname{tr}(P_{(2,2,1)}R_5P_{(2,2,1)}\Pi)\\ &= (1+q)\operatorname{tr}(P_{(2,2,1)}D_5R_5'P_{(2,2,1)}\Pi) + v\operatorname{tr}(P_{(2,2,1)}M_5R_5'P_{(2,2,1)}\Pi).\end{aligned}$$

For the first summand: since $V_{(2,2,1)} \subseteq \text{ascents}_5$, the diagonal D_5 acts as the identity on $V_{(2,2,1)}$, so $P_{(2,2,1)}D_5 = P_{(2,2,1)}$. Hence

$$\operatorname{tr}(P_{(2,2,1)}D_5R_5'P_{(2,2,1)}\Pi) \;=\; \operatorname{tr}(P_{(2,2,1)}R_5'P_{(2,2,1)}\Pi) \;=\; \operatorname{cross}^{KL}_{(2,2,1)\to(2,2,1)}(R_5'\Pi).$$

For the second summand: M_5 has rows in s_5-descent vertices, but $V_{(2,2,1)} \subseteq \text{ascents}_5$, so $P_{(2,2,1)}M_5 = 0$. Hence the second summand vanishes.

Combining, $\operatorname{cross}^{KL}_{(2,2,1)\to(2,2,1)}(R_5\Pi) = (1+q)\operatorname{cross}^{KL}_{(2,2,1)\to(2,2,1)}(R_5'\Pi)$.

Equality of X and $\operatorname{cross}^{KL}_{(2,2,1)\to(2,2,1)}(R_5'\Pi)$. Numerically these are equal: both equal $q^6(1+q)^2$. The structural reason is the observation that $R_5'\Pi_q^{S_5} \in H_q(S_5)$ has zero off-block "flux" from non-$(2,2,1)$ to $(2,2,1)$ in the KL basis, i.e.,

$$\sum_{w\in V_{(2,2,1)},\, l\in V_{(3,2)}\cup V_{(3,1,1)}} (R_5')_{wl}\,(\Pi_q^{S_5})_{lw} \;=\; 0 \tag{V3}$$

identically in q. We give Identity (V3) the status of a precise structural gap: it is verified numerically by the data

$$X = q^6(1+q)^2 = \operatorname{cross}^{KL}_{(2,2,1)\to(2,2,1)}(R_5'\Pi) \quad \text{(numerical)}.$$

A structural proof of (V3) would reduce Lemma 2 to a fully structural statement. □

Step 3: identifying $(1+q)Y$ as a cross-block trace

Lemma 3 (Diagonal cross-trace at $V_{(3,1,1)}$)**.**

$$\operatorname{cross}^{KL}_{(3,1,1)\to(3,1,1)}(R_5\,\Pi_q^{S_5}) \;=\; (1+q)\,Y.$$

Proof. By the same expansion as in Lemma 2:

$$\operatorname{cross}^{KL}_{(3,1,1)\to(3,1,1)}(R_5\Pi) = (1+q)\operatorname{tr}(P_{(3,1,1)}D_5R_5'P_{(3,1,1)}\Pi) + v\operatorname{tr}(P_{(3,1,1)}M_5R_5'P_{(3,1,1)}\Pi).$$

For the first summand: $V_{(3,1,1)}$-block has descent subset $\{6,12,14\}$ and ascent subset $\{1,3,9\}$. D_5 projects to ascents, so $P_{(3,1,1)}D_5 = P_{\{1,3,9\}}$. Hence

$$\operatorname{tr}(P_{(3,1,1)}D_5R_5'P_{(3,1,1)}\Pi) \;=\; \operatorname{tr}(P_{\{1,3,9\}}R_5'P_{(3,1,1)}\Pi) \;=:\; Y'.$$

For the second summand: $P_{(3,1,1)}M_5P_{(3,1,1)}$ has rows in $\{6,12,14\}$ (descent subset of $V_{(3,1,1)}$) and columns in $\{1,3,9\}$ (ascent subset). This is non-zero in general. So the M-piece contributes a non-trivial

$$Q \;:=\; \operatorname{tr}(P_{(3,1,1)}M_5R_5'P_{(3,1,1)}\Pi).$$

Combining, $\operatorname{cross}^{KL}_{(3,1,1)\to(3,1,1)}(R_5\Pi) = (1+q)Y' + vQ$.

The claim of the Lemma is that this equals $(1+q)Y$. Equivalently:

$$(1+q)(Y-Y') \;=\; vQ. \tag{1}$$

Now Y and Y' differ only in the range of summation of the second index: Y sums over all $l \in V_\lambda$, Y' sums over $l \in V_{(3,1,1)}$. So

$$Y - Y' \;=\; \sum_{w\in\{1,3,9\},\, l\in V_{(3,2)}\cup V_{(2,2,1)}} (R'_5)_{wl}\,(\Pi_q^{S_5})_{lw}.$$

Equation (1) is verified numerically. A structural proof would identify the precise mechanism by which the M_5-coupling Q captures the "spillover" $Y - Y'$ between $V_{(3,1,1)}$ and the other two blocks. We identify two precise sub-statements:

(V1) $\sum_{w\in\{6,12,14\}} (R'_5\,\Pi_q^{S_5})_{ww} \;=\; 0$ identically — the diagonal trace of $R'_5\Pi$ on the s_5-descent subset of $V_{(3,1,1)}$ is zero.

(V2) $\mathrm{tr}(P_{(3,2)}\,R'_5\,P_{(3,1,1)}\,\Pi_q^{S_5}) = 0$ identically — the cross-block trace from $V_{(3,1,1)}$ to $V_{(3,2)}$ via $R'_5\Pi$ (i.e., *without* a T_5+1 factor) vanishes.

Both are verified numerically (see Section 3). They constitute the structural gap of the Lemma. □

Step 4: the main decomposition

Theorem 4 (Main decomposition; partly conditional)**.** *Modulo the conjectures* (V3) *and* (V_Y) *(Proposition 11):*

$$D(q) \;=\; \mathrm{cross}^{KL}_{(2,2,1)\to(2,2,1)}(R_5\,\Pi_q^{S_5}) \;+\; \mathrm{cross}^{KL}_{(3,1,1)\to(3,1,1)}(R_5\,\Pi_q^{S_5}).$$

Equivalently, the D-piece of the Q3 split equals the sum of the two diagonal cross-block traces at $V_{(2,2,1)}$ and $V_{(3,1,1)}$.

Proof. Combine Lemma 1 (structural) with Lemmas 2 (modulo (V3)) and 3 (modulo (V_Y)):

$$\begin{aligned}
\mathrm{cross}^{KL}_{(2,2,1)\to(2,2,1)}(R_5\Pi) + \mathrm{cross}^{KL}_{(3,1,1)\to(3,1,1)}(R_5\Pi) &= (1+q)X + (1+q)Y \\
&= (1+q)(X+Y) \\
&= D(q).
\end{aligned}$$

The first equality uses Lemmas 2 and 3; the second is linearity; the third is Lemma 1, which is unconditional. □

Remark 5 (The unconditional content)**.** What is fully proved structurally — without any conjectures — is Lemma 1:

$$D(q) \;=\; (1+q)\cdot[X+Y] \;=\; (1+q)\cdot \mathrm{tr}\big(P_{\mathrm{ascents}_5}\,R'_5\,\Pi_q^{S_5}\big).$$

This is the structural reformulation: $D(q)$ is exactly $(1+q)$ times the diagonal trace of $R'_5\Pi_q^{S_5}$ on the s_5-ascent set $V_{(2,2,1)}\cup\{1,3,9\}$. From this and the numerical observations $X = q^6(1+q)^2$ and $Y = q^5(1+q)^2(1+4q+q^2)$, one obtains the value of $D(q)$ directly. The further identification of X, Y with KL diagonal cross-block traces (and hence Theorem 4) requires the conjectures named above.

Step 5: the surprising identity, restated

Corollary 6 (Trace identity, equivalent form)**.** *The trace identity* $D(q) = \mathrm{cross}^{KL}_{(3,1,1)\to(3,2)}(R_5\Pi_q^{S_5})$ *is equivalent (modulo Theorem 4) to the three-way KL cross-trace identity*

$$\mathrm{cross}^{KL}_{(2,2,1)\to(2,2,1)}(R_5\Pi) \;+\; \mathrm{cross}^{KL}_{(3,1,1)\to(3,1,1)}(R_5\Pi) \;=\; \mathrm{cross}^{KL}_{(3,1,1)\to(3,2)}(R_5\Pi).$$

Proof. Direct substitution into the surprising identity using Theorem 4. □

Remark 7 (Why this reformulation is interesting)**.** The right-hand side $\mathrm{cross}^{KL}_{(3,1,1)\to(3,2)}$ is an *off-diagonal* cross-block trace: a sum over matrix entries with indices in two different μ-blocks. The left-hand side is a sum of two *diagonal* cross-block traces. These are very different structural objects. Their numerical equality is a nontrivial identity among integer polynomials in q.

The reformulation does not yet identify a single combinatorial bijection between paths counted by either side, but it isolates the question: why does the "diagonal-block contribution from $V_{(2,2,1)}$ and $V_{(3,1,1)}$" equal the "off-diagonal contribution from $V_{(3,1,1)}$ to $V_{(3,2)}$"?

3 The structural gap: identities (V1), (V2), (V3)

The proof of Theorem 4 above is reduced to three numerical identities. We restate them precisely.

Conjecture 8 (V1: descent-block diagonal vanishing)**.** $\sum_{w\in\{6,12,14\}}(R'_5\,\Pi_q^{S_5})_{ww} \;=\; 0$ identically in $\mathbb{Z}[q]$.

Conjecture 9 (V2: cross-block R'_5-trace vanishing)**.** $\mathrm{tr}\big(P_{(3,2)}\,R'_5\,P_{(3,1,1)}\,\Pi_q^{S_5}\big) \;=\; 0$ identically.

Conjecture 10 (V3: $V_{(2,2,1)}$ off-block R'_5-flux vanishing)**.** $\sum_{w\in V_{(2,2,1)},\, l\notin V_{(2,2,1)}}(R'_5)_{wl}\,(\Pi_q^{S_5})_{lw} \;=\; 0$ identically.

Each is verified numerically by direct computation in `~/projects/scratch/2026-05-06-trace-identity/` (May 6 evening). We prove that the three identities, taken together, are sufficient for the main Theorem.

Proposition 11 (Partial sufficiency)**.** *Conjecture 10 alone (the $V_{(2,2,1)}$ off-block R'_5-flux vanishing) suffices to make Lemma 2 fully structural:* $\mathrm{cross}^{KL}_{(2,2,1)\to(2,2,1)}(R_5\Pi) = (1+q)X$.

The full Theorem 4 requires, in addition, the further structural identity

$$Y \;=\; \mathrm{tr}\big(P_{\{1,3,9\}}\,R'_5\,P_{V_{(3,1,1)}}\,\Pi_q^{S_5}\big) \;+\; vQ/(1+q), \qquad (\mathrm{V}_Y)$$

where $Q = \mathrm{tr}(P_{(3,1,1)}M_5R'_5P_{(3,1,1)}\Pi)$ *is the* M_5*-mediated cross-term within* $V_{(3,1,1)}$*. This identity is verified numerically by the data* (11) $= (1+q)Y$*, but its structural proof is not contained in (V1), (V2), (V3) alone.*

Proof. $V_{(2,2,1)}$ **piece (clean).** Lemma 2's structural step gave $\mathrm{cross}^{KL}_{(2,2,1)\to(2,2,1)}(R_5\Pi) = (1+q)\,\mathrm{cross}^{KL}_{(2,2,1)\to(2,2,1)}(R'_5\Pi)$, using only the fact that $V_{(2,2,1)} \subseteq \mathrm{ascents}_5$. Conjecture (V3) then gives $\mathrm{cross}^{KL}_{(2,2,1)\to(2,2,1)}(R'_5\Pi) = X$, hence $(1+q)X$.

$V_{(3,1,1)}$ **piece (gap).** Lemma 3's structural derivation gave

$$\mathrm{cross}^{KL}_{(3,1,1)\to(3,1,1)}(R_5\Pi) \;=\; (1+q)Y' + vQ,$$

where $Y' := \mathrm{tr}(P_{\{1,3,9\}}R'_5P_{(3,1,1)}\Pi)$ (sum restricted to $l \in V_{(3,1,1)}$) and Q is the M_5-mediated within-block cross-term.

The claim $\mathrm{cross}_{(3,1,1)\to(3,1,1)}(R_5\Pi) = (1+q)Y$ requires $(1+q)Y' + vQ = (1+q)Y$, i.e., $vQ = (1+q)(Y-Y')$. Now

$$Y - Y' \;=\; \sum_{w\in\{1,3,9\},\, l\in V_{(3,2)}\cup V_{(2,2,1)}} (R_5')_{wl}\,\Pi_{lw},$$

which involves the cross-block flow from $\{1,3,9\}$ to non-$V_{(3,1,1)}$ via $R_5'\Pi$. Conjectures (V1), (V2), (V3) do not directly express this quantity. The relation $vQ = (1+q)(Y-Y')$ is a separate structural statement; we record it as a sub-conjecture (V_Y) of the same flavor. Numerically, $vQ = (1+q)(Y-Y') = q^6 + q^7 + 6q^8 + 6q^9 + q^{10} + q^{11}$ (computable from (11) $-\,(1+q)Y' = vQ$), but a structural proof is missing.

So (V3) alone closes the $V_{(2,2,1)}$ identification, but the $V_{(3,1,1)}$ identification remains conditional on (V_Y) (or any structural argument linking vQ to the cross-flow $Y - Y'$). □

Remark 12 (Provability status)**.** We have not proven (V1), (V2), or (V3) structurally. They are stated as conjectures, supported by exact numerical computation. Each is a precise KL-basis trace identity for the operator $R_5'\,\Pi_q^{S_5} \in H_q(S_5)$ acting on $V_{(3,2,1)}$ at S_6.

The structural meaning of these identities likely involves the S_4-double-branching of $V_{(3,2,1)}$, where $V_{(3,2,1)} \downarrow S_4 = 2V_{(3,1)} \oplus 2V_{(2,2)} \oplus 2V_{(2,1,1)}$. In each isotypic 2-fold component $V_\nu^{(\lambda)}$, the T_5-action is $I_{V_\nu} \otimes A_\nu$ for some 2×2 matrix A_ν on the multiplicity space, and the descent/ascent split of the multiplicity space matches the V_μ-block structure: one copy is s_5-descent (in V_μ for some specific μ) and one is s_5-ascent (in another V_μ). The vanishing identities (V1)–(V3) likely express specific coupling constraints in this structure.

4 The S_4-double-branching framework

We sketch the structural framework that suggests where (V1), (V2), (V3) come from, even though we cannot yet close the gap.

S_4-double-branching of $V_{(3,2,1)}$

The Young lattice gives $V_{(3,2,1)} \downarrow S_4 = \bigoplus_\nu m_\nu\, V_\nu^{S_4}$ where ν ranges over partitions of 4 obtained by removing two corner boxes from $(3,2,1)$, with multiplicity m_ν the number of paths in the lattice. We computed:

- $\nu = (3,1)$: paths $(3,2,1) \to (3,2) \to (3,1)$ and $(3,2,1) \to (3,1,1) \to (3,1)$. $m_{(3,1)} = 2$.
- $\nu = (2,2)$: paths via $(3,2)$ and $(2,2,1)$. $m_{(2,2)} = 2$.
- $\nu = (2,1,1)$: paths via $(3,1,1)$ and $(2,2,1)$. $m_{(2,1,1)} = 2$.

Hence $V_{(3,2,1)} \downarrow S_4 = 2V_{(3,1)} \oplus 2V_{(2,2)} \oplus 2V_{(2,1,1)}$ of dimensions $6+4+6 = 16$. ✓

T_5 commutes with $H_q(S_4)$

Since $|5-i| \geq 2$ for $i \leq 3$, T_5 commutes with T_1, T_2, T_3 — i.e., with $H_q(S_4)$. Hence T_5 preserves the S_4-isotypic decomposition above. On each $V_\nu^{(\lambda)} \cong V_\nu^{S_4} \otimes \mathbb{C}^{m_\nu = 2}$,

$$T_5\big|_{V_\nu^{(\lambda)}} \;=\; I_{V_\nu^{S_4}} \otimes A_\nu, \qquad A_\nu \in \mathrm{End}(\mathbb{C}^2) \otimes \mathbb{Z}[q,v].$$

Each A_ν has eigenvalues $\{q, -1\}$. The $\mathbb{C}^2$ basis is indexed by the two paths $\lambda \to \mu \to \nu$, and the descent/ascent assignment is

- $\nu = (3,1)$: descent path is $(3,2,1) \to (3,2) \to (3,1)$ (5 above 6, axial distance $+2$); ascent path is $(3,2,1) \to (3,1,1) \to (3,1)$ (5 below 6, axial distance -2).
- $\nu = (2,2)$: descent path is $(3,2,1) \to (3,2) \to (2,2)$ (axial distance $+4$); ascent path is $(3,2,1) \to (2,2,1) \to (2,2)$ (axial distance -4).
- $\nu = (2,1,1)$: descent path is $(3,2,1) \to (3,1,1) \to (2,1,1)$ (axial distance $+2$); ascent path is $(3,2,1) \to (2,2,1) \to (2,1,1)$ (axial distance -2).

By Hoefsmit's seminormal formula, the diagonal entries of A_ν in this basis are

$$A_\nu^{\text{descent}} = -\frac{1}{[d]_q}, \qquad A_\nu^{\text{ascent}} = \frac{q^d}{[d]_q},$$

where d = axial distance ($d = 2$ for $\nu \in \{(3,1),(2,1,1)\}$ and $d = 4$ for $\nu = (2,2)$), and $[d]_q = (q^d - 1)/(q-1)$.

Cross-block flow via S_4-DB

The S_4-DB structure interlocks the S_5-block decomposition: each isotypic $V_\nu^{(\lambda)}$ has its descent copy in one S_5-block and its ascent copy in another. Specifically:

- $V_{(3,1)}^{(\lambda),d} \subset V_{(3,2)}$, $V_{(3,1)}^{(\lambda),a} \subset V_{(3,1,1)}$.
- $V_{(2,2)}^{(\lambda),d} \subset V_{(3,2)}$, $V_{(2,2)}^{(\lambda),a} \subset V_{(2,2,1)}$.
- $V_{(2,1,1)}^{(\lambda),d} \subset V_{(3,1,1)}$, $V_{(2,1,1)}^{(\lambda),a} \subset V_{(2,2,1)}$.

The $V_{(3,2)}$-block consists of $V_{(3,1)}^{(\lambda),d} \oplus V_{(2,2)}^{(\lambda),d}$ ($3+2=5$ vertices). The $V_{(3,1,1)}$-block consists of $V_{(2,1,1)}^{(\lambda),d} \oplus V_{(3,1)}^{(\lambda),a}$ ($3+3=6$ vertices). The $V_{(2,2,1)}$-block consists of $V_{(2,1,1)}^{(\lambda),a} \oplus V_{(2,2)}^{(\lambda),a}$ ($3+2=5$ vertices).

In particular, the s_5-descent subset $\{6,12,14\}$ of $V_{(3,1,1)}$ is exactly $V_{(2,1,1)}^{(\lambda),d}$ (the descent copy of $V_{(2,1,1)}$).

Why (V1) is plausible

(V1) says $\mathrm{tr}_{V_{(2,1,1)}^{(\lambda),d,KL}}(R_5' \Pi_q^{S_5}) = 0$, where the trace is over the KL-basis vectors lying in the descent copy of $V_{(2,1,1)}$ inside $V_{(3,1,1)}$.

In the seminormal basis adapted to the S_4-DB, the descent copy and the ascent copy of $V_{(2,1,1)}$ are linked by the (basis-independent) operator $T_5 + 1$, which has a specific non-zero off-block component. The KL-basis projector onto $V_{(2,1,1)}^{(\lambda),d,KL}$ is (in general) not the same as the seminormal $H_q(S_4)$-equivariant projector.

(V1)'s vanishing is the statement that the KL-basis sum behaves as if there's an $H_q(S_4)$-equivariance breaking that "averages out" the trace in a specific way. A complete proof would require explicit description of the change of basis between KL and seminormal on $V_{(3,2,1)}$, which we have not yet computed.

5 Computational verification

All polynomial values used in this paper are computed in `~/projects/scratch/2026-05-06-trace-identity/com` (May 6, evening), using the W-graph data from `~/projects/scratch/2026-05-06-paths/` via Lagrange interpolation on integer evaluations of the matrix product. The polynomials reported in the abstract and Section 1 are the exact values, verified by independent recomputation. All cross-checks (sums to σ_1, palindromicity, factorization) pass.

6 Open questions

1. **Structural proof of (V1), (V2), (V3).** Each is a numerically verified identity in $\mathbb{Z}[q]$, expressing a specific KL-basis cross-block trace vanishing. A proof would likely use: (a) the explicit S_4-double-branching change-of-basis from KL to seminormal, or (b) a W-graph structural argument about $R_5' \Pi_q^{S_5}$.
2. **Generalization.** The $D(q) = \mathrm{cross}^{KL}_{(3,1,1)\to(3,2)}$ identity at $\lambda = (3,2,1)$ — does an analogous identity hold for other self-conjugate or near-self-conjugate λ at S_n? Specifically, for which $\lambda \vdash n$ is the D-piece (the trace contribution where the s_{n-1} step uses $(1+q)D_{n-1}$) equal to a specific cross-block trace?
3. **Path-level interpretation.** The D-piece counts closed W-graph paths in $V_{(3,2,1)}$ where step 11 is a D_5-step. The cross-block counts paths whose start-end blocks differ. The numerical equality suggests a bijection between the two sets of paths. We have not exhibited such a bijection.

Honest assessment

The contribution of this writeup is the structural decomposition of $D(q)$ into a sum of two diagonal cross-block traces (Theorem 4), and the precise identification of the gap as three numerically verified KL-basis trace identities (V1), (V2), (V3). What is gained:

- The surprising identity is restated as a clean three-way relation among KL cross-traces, which is more amenable to structural attack than the original "D = cross" formulation.
- The role of the s_5-ascent/descent splitting of $V_{(3,1,1)}$ becomes visible — the descent subset $\{6, 12, 14\}$ contributes zero, the ascent subset $\{1, 3, 9\}$ contributes the full Y.
- The $V_{(2,2,1)}$ piece is fully structural (Lemma 2 modulo (V3)).

What is not gained:

- A proof of (V1), (V2), (V3) themselves. These remain as open structural gaps.
- A path-level bijection (the level we ultimately want).

This writeup is logged as a precise reduction: the path-level Theorem B and the trace identity decomposition both now have specific, named structural gaps. The author flags the present result as a reduction, not a closure.